

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

May 2014

EE310 Programmable Logic Controller and Industrial Automation

Date: 15.05.2014, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

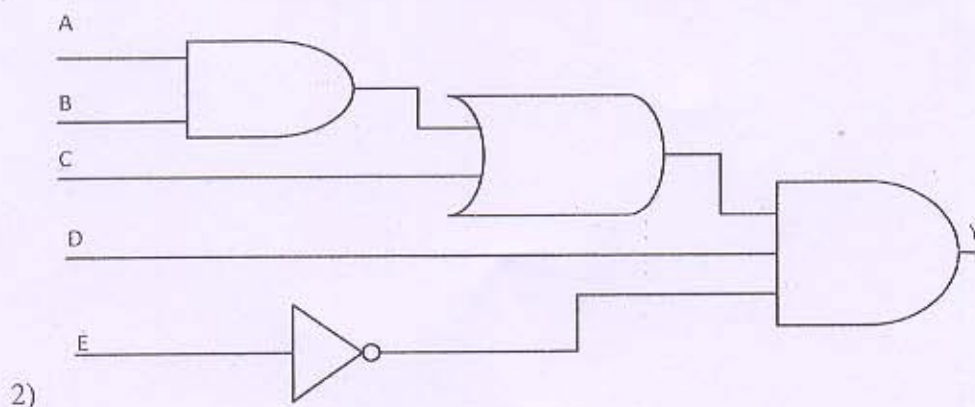
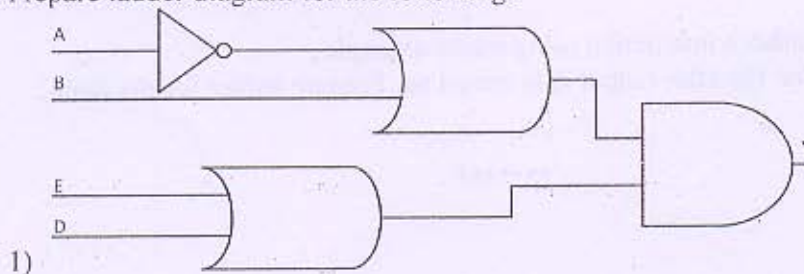
Q 1 Explain difference between sequential and analog control. [07]

Q 2(a) Explain block diagram of PLC. [07]

(b) Explain function and classification of input module. [07]

OR

Q 2 Prepare ladder diagram for the following: [14]



Q3 (a) Explain advantages and disadvantages of PLC. [07]

(b) Explain evolution of control system. [07]

OR

Q3 Implement AND, OR, NOT, NAND, NOR, EXOR and EXNOR gates in ladder logic. [14]

SECTION – II

- Q4 Give classification of various types of timers and explain TON timer in detail. [07]
- Q5(a) There are 4 outputs R, S, T and U. R starts immediately when an input is energized. S starts 4s later. T starts 5s later than S. U go on 1.9s after S. One switch turns all outputs off. Prepare ladder diagram for the same. [07]
- (b) A fan F is to be turned on when count L goes from 7 down to 0 and when either count M goes up to 14 or count N has not gone all the way from 14 down to 0. One switch or stop button resets the entire process. Prepare ladder for the same. [07]

OR

- Q5(a) State the 5 names of 5 types of programming language used in PLC. [05]
- (b) Explain scan cycle of PLC. [07]
- (c) Draw symbol of Normally open and normally closed instruction. [02]
- Q6(a) Explain TOFF timer in detail. [05]
- (b) Classify counter and explain up counter in detail. [05]
- (c) Explain down counter in detail. [04]

OR

- Q6(a) Explain latch and unlatch instruction using motor example. [07]
- (b) Output B goes on for 10s after output A is turned on. Prepare ladder for the same. [07]
